Cross-Domain Recommendation under MF: Full Comparison

Seed: 2020. Three tasks; $\beta \in \{20\%, 50\%, 80\%\}$; six methods.

Table 1: MAE on three cross-domain tasks (MF, seed 2020). Lower is better.

Task	β	TGTOnly	EMCDR	PFCDR	LIST V1	LIST V1 + Langevin	LAPT
Book → Movie	20%	4.2695	1.1579	0.9498	0.8148	0.8123	**0.8107**
	50%	4.2261	1.2444	1.0054	0.8207	**0.8192**	0.8198
	80%	4.2746	1.3524	1.1076	0.8361	**0.8360**	0.8363
Book → Music	20%	4.4361	1.6416	1.4804	0.7148	**0.7126**	0.7141
	50%	4.5043	1.8280	1.8385	**0.7304**	0.7314	0.7314
	80%	4.5540	2.1528	2.3173	**0.7296**	0.7299	0.7297
Music → Movie	20%	4.1979	1.2641	0.9997	0.8250	0.8224	**0.8178**
	50%	4.1829	1.2595	1.0155	0.8363	**0.8342**	0.8365
	80%	4.2371	1.2886	1.0472	**0.8392**	0.8407	0.8407

Table 2: RMSE on three cross-domain tasks (MF, seed 2020). Lower is better.

Task	β	TGTOnly	EMCDR	PFCDR	LIST V1	LIST V1 + Langevin	LAPT
Book → Movie	20%	4.8407	1.4581	1.2211	1.0644	**1.0624**	1.0644
	50%	4.8444	1.5563	1.2996	**1.0756**	1.0761	1.0760
	80%	4.9229	1.6994	1.4449	**1.0994**	1.1001	1.1001
Book → Music	20%	5.1502	1.9526	1.7424	0.9679	0.9674	**0.9672**
	50%	5.2445	2.1701	2.1219	**0.9970**	0.9971	0.9971
	80%	5.3280	2.5241	2.6122	**0.9956**	0.9956	0.9956
Music → Movie	20%	4.7979	1.6434	1.2753	1.0717	1.0703	**1.0694**
	50%	4.7968	1.6361	1.3067	1.0861	**1.0848**	1.0864
	80%	4.8683	1.6872	1.4005	**1.0971**	1.0976	1.0976

Protocol. β is the fraction of overlapping users held out as target cold-start test users. We retain all target-only users in training. Music-to-Movie removes cross-domain duplicate items from the source and contains 17,904 overlapping users (the original paper has 18,031). The other tasks contain 37,388 and 16,738 overlapping users.

Backbone and training. TGTOnly, EMCDR and PFCDR use the official 10-dimensional MF. All LIST variants receive the same newly trained MF Top-100 IDs/ranks; their established 32-dimensional list/target representations are retained. Source and target MF train for 10 epochs using Adam (0.01). Source/target batches are 512 for Book tasks and 256 for Music-to-Movie. EMCDR uses mapping batch 128 and learning rate 0.01. PFCDR uses full-catalog inversion (300 iterations, batch 200), mapping learning rate 0.005, mapping batch 512/512/1024, and the released MF implementation's 70,000-user voting rule.

LIST variants. LIST V1 uses 2,048 interaction batches; LIST V1 + Langevin and LAPT use 64 complete-user batches. Both Langevin variants retain the same fixed sampling settings. LAPT adds the current rank-aware shared-pattern reconstruction and residual attention with masked-ID supervision. No SID/coarse codes or additional hyperparameter search are used.

Evaluation. Epochs (and LIST correction weights) are selected on an independent inner validation split, then models are freshly refit on all outer training users. All 54 prediction arrays are frozen before test-label evaluation. This differs from the released code's best-test-epoch reporting. Results are for one seed, not the paper's ten-run averages. Bold identifies the best unrounded value within each row. No statistical-significance claim is made.

Mean fallback. Validation selects $\lambda = 0$ in 3 of nine LAPT settings and 2 of nine LIST V1 + Langevin settings; these predictions equal ordinary training-item means. LIST V1 selects nonzero corrections in all nine settings.